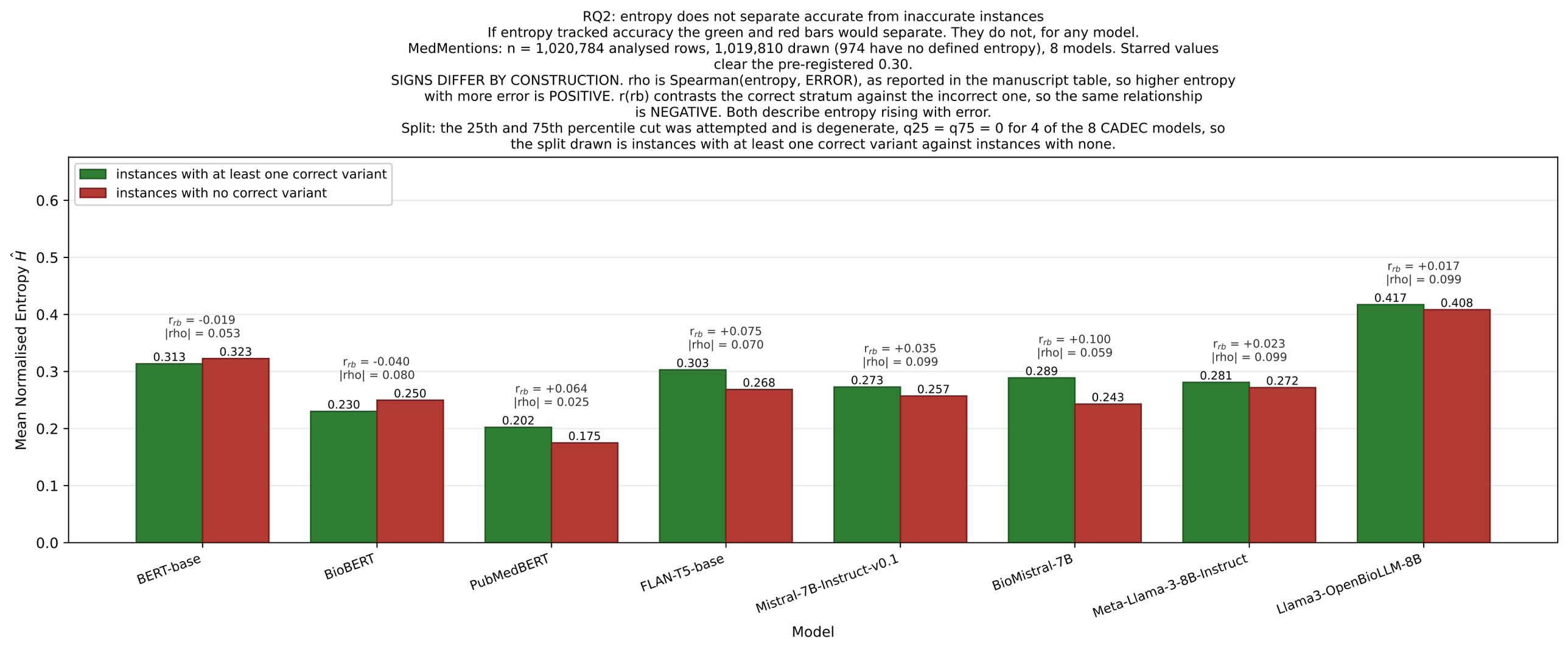
RQ2: entropy does not separate accurate from inaccurate instances
If entropy tracked accuracy the green and red bars would separate. They do not, for any model.
MedMentions: n = 1,020,784 analysed rows, 1,019,810 drawn (974 have no defined entropy), 8 models. Starred values clear the pre-registered 0.30.
SIGNS DIFFER BY CONSTRUCTION. rho is Spearman(entropy, ERROR), as reported in the manuscript table, so higher entropy with more error is POSITIVE. r(rb) contrasts the correct stratum against the incorrect one, so the same relationship is NEGATIVE. Both describe entropy rising with error.
Split: the 25th and 75th percentile cut was attempted and is degenerate, q25 = q75 = 0 for 4 of the 8 CADEC models, so the split drawn is instances with at least one correct variant against instances with none.
instances with at least one correct variant
instances with no correct variant
Mean Normalised Entropy $\hat{H}$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
r_{rb} = -0.019
|rho| = 0.053
0.313
0.323
BERT-base
r_{rb} = -0.040
|rho| = 0.080
0.230
0.250
BioBERT
r_{rb} = +0.064
|rho| = 0.025
0.202
0.175
PubMedBERT
r_{rb} = +0.075
|rho| = 0.070
0.303
0.268
FLAN-T5-base
r_{rb} = +0.035
|rho| = 0.099
0.273
0.257
Mistral-7B-Instruct-v0.1
r_{rb} = +0.100
|rho| = 0.059
0.289
0.243
BioMistral-7B
r_{rb} = +0.023
|rho| = 0.099
0.281
0.272
Meta-Llama-3-8B-Instruct
r_{rb} = +0.017
|rho| = 0.099
0.417
0.408
Llama3-OpenBioLLM-8B
Model